

1. 2. 1. 1.

1. kontrolna naloga
3. A, 9. 10. 2024

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Čas pisanja: 45 min



dosežene točke	možne točke	odstotki	ocena
30	36	83	4

1. a) Določi funkciji $f(x) = 2^{2x-4} + 1$ inverzno funkcijo f^{-1} .

[4t] 4

$$\begin{aligned}
 y &= 2^{2x-4} + 1 \\
 x &= 2^{2x-4} + 1 \\
 x-1 &= 2^{2x-4} \\
 \log_2(x-1) &= 2x-4 \\
 \log_2(x-1) &= 2x-4 \\
 x-1 &= 4^{x-2} \\
 \log_4(x-1) &= \log_4 4^{x-2} \\
 x-1 &= 4^{x-2} \\
 \log_4(x-1) &= \log_4 4^{x-2} \\
 y-2 &= \log_4(x-1) \\
 y &= \log_4(x-1) + 2 = f^{-1}(x)
 \end{aligned}$$

- b) Nariši graf funkcije $g(x) = 2^{|2x-4|} + 1$.

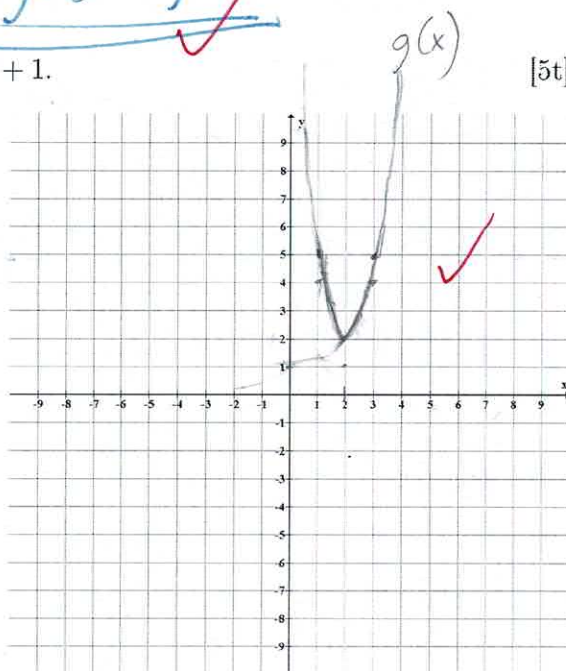
[5t] 5

$$2x-4 \geq 0 \Rightarrow x \geq 2$$

$$\begin{aligned}
 g(x) &= 2^{2(x-2)} + 1 = \\
 &= 4^{x-2} + 1 \quad \checkmark
 \end{aligned}$$

$$2x-4 < 0 \Rightarrow x < 2$$

$$g(x) = \left(\frac{1}{4}\right)^{x-2} + 1 \quad \checkmark$$



2. Reši enačbi:

a) $10 \cdot 2^{2x+1} - 9^{x+2} = 23 \cdot 3^{2x+1} - 5 \cdot 4^{x+2}$

[5t] 5

$$2 \cdot 10 \cdot 2^{2x+1} - 23 \cdot 3^{2x+1} \cdot 3 = 9^{x+2} - 5 \cdot 4^{x+2} \cdot 4^2$$

$$20 \cdot 4^x - 23 \cdot 9^x \cdot 3 = 81 \cdot 9^x - 80 \cdot 4^x$$

$$20 \cdot 4^x + 80 \cdot 4^x = 81 \cdot 9^x + 69 \cdot 9^x$$

$$100 \cdot 4^x = 150 \cdot 9^x$$

$$\left(\frac{9}{4}\right)^x = \frac{2}{3}$$

$$\frac{100}{150} = \frac{9^x}{4^x}$$

$$\left(\frac{9}{4}\right)^x = \frac{100}{150} \cdot \frac{2}{3}$$

$$x = -\frac{1}{2}$$

$$\log_{\frac{9}{4}} \frac{100}{150} = x = \log_{\frac{9}{4}} \frac{2}{3} = -\frac{1}{2}$$

b) $\log_{\frac{1}{2}} x^2 - \log_8 \frac{1}{x} + \log_4 x + \log_{16} x = \frac{11}{12}$

[5t] 5

$$\frac{\log_2 x^2}{\log_2 \frac{1}{2}} - \frac{\log_2 \frac{1}{x}}{\log_2 8} + \frac{\log_2 x}{\log_2 4} + \frac{\log_2 x}{\log_2 16} = \frac{11}{12}$$

$$\frac{\log_2 x^2}{-1} - \frac{\log_2 1 - \log_2 x}{3} + \frac{\log_2 x}{2} + \frac{\log_2 x}{4} = \frac{11}{12} \quad / \cdot 12$$

$$-12 \log_2 x^2 - 12 \cdot \left(\frac{0 - \log_2 x}{3}\right) + 6 \log_2 x + 3 \log_2 x = 11$$

$$-24 \log_2 x + 4 \log_2 x + 6 \log_2 x + 3 \log_2 x = 11$$

$$-24 \log_2 x + 13 \log_2 x = 11$$

$$-11 \log_2 x = 11$$

$$\log_2 x = -1$$

$$x = \frac{1}{2}$$

P: L: $\log_{\frac{1}{2}} \left(\frac{1}{2}\right)^2 - \log_8 \frac{1}{\frac{1}{2}} + \log_4 \frac{1}{2} + \log_{16} \frac{1}{2} =$

$$= 2 - \log_8 2 + \frac{1}{2} + \frac{1}{4} = 2 + \frac{1}{3} - \frac{2}{4} - \frac{1}{4} = 2 + \frac{4}{12} - \frac{3}{4} =$$

$$= 2 + \frac{4}{12} - \frac{9}{12} = \frac{11}{12}$$

D: $\frac{11}{12}$ ✓

3. Podani sta funkciji $f(x) = \ln(2x - 7)$ in $g(x) = 2\ln x - \ln d$, $d \in \mathbb{R}^+$.

a) Zapiši enačbo navpične asimptote grafa funkcije f .

[2t] 2

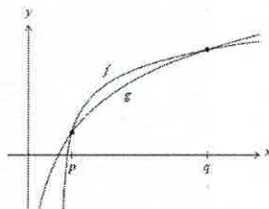
$$\ln(2x-7) \quad 2x-7=0 \quad 2x=7 \quad x=\frac{7}{2} \quad \checkmark$$

Grafa $y = f(x)$ in $y = g(x)$ se sekata v dveh različnih točkah. Abscisi presečišč sta $x = p$ in $x = q$, kjer je $p < q$.

$$f(p) = g(p)$$

$$f(q) = g(q)$$

$$f(p) = g(p)$$



b) Naj bo $d = 10$. Izračunaj vrednost $q - p$ in rezultat zapiši v obliki $a\sqrt{b}$, pri čemer sta $a, b \in \mathbb{Z}^+$.

[5t] 3

$$\ln(2x-7) = 2\ln x - \ln 10$$

$$\ln(2x-7) = \ln\left(\frac{x^2}{10}\right)$$

$$2x-7 = \frac{x^2}{10}$$

$$20x-70 = x^2$$

$$-x^2 + 20x - 70 = 0$$

$$x^2 - 20x + 70 = 0$$

$$x = \frac{20 \pm \sqrt{400 - 56}}{2} = \frac{20 \pm \sqrt{344}}{2}$$

$$p = \frac{20 - \sqrt{344}}{2}$$

$$q = \frac{20 + \sqrt{344}}{2}$$

$$q - p = \frac{20 + \sqrt{344}}{2} - \frac{20 - \sqrt{344}}{2} = \frac{2\sqrt{344}}{2} = \sqrt{344} = 2\sqrt{86}$$

c) Pokaži, da za absciso presečišča grafov velja enačba $x^2 - 2dx + 7d = 0$.

[2t] 2

$$f(x) = g(x)$$

$$\ln(2x-7) = \ln\left(\frac{x^2}{d}\right)$$

$$2x-7 = \frac{x^2}{d}$$

$$x^2 = 2dx - 7d \Rightarrow x^2 - 2dx + 7d = 0$$

č) Določi vse možne vrednosti d .

[3t] 0

$$d(2x+7) = -x^2 \quad d > 0$$

$$d = \frac{-x^2}{2x+7} \quad \frac{-x^2}{2x+7} > 0$$

$$2x+7 < 0$$

$$x < -\frac{7}{2}$$

d so vsa $\mathbb{R}^+$, ko je $x < -\frac{7}{2}$, drugače tak d ne obstoje.

4. Reši neenačbo: $\log_2(x-3) + \log_2 \frac{1}{x-1} < 1$.

[5t] 3

$$\log_2(x-3) \left(\frac{1}{x-1} \right) < 1$$

$$\log_2 \left(\frac{x-3}{x-1} \right) < 1 \quad / : 2^n$$

$$2^{\log_2 \left(\frac{x-3}{x-1} \right)} < 2$$

$$\frac{x-3}{x-1} < 2$$

$$\rightarrow \forall x > 3$$

$$x > 1$$

$$\cancel{x > 1}$$

$$\underline{x > 1}$$

$$R: x > 3$$

$$\underline{x < 1}$$

ima ~~pa~~ potencialen
negativni logaritmand

$$\underline{x > 1}$$

$$x-3 < 2(x-1)$$

$$x-3 < 2x-2$$

$$-x < 1$$

$$\cancel{x < -1} \quad x > -1$$

$$\underline{x = 1}$$

Deljenje z 0

DODATNA NALOGA:

Naj bo f funkcija, ki naravnemu številu priredi število števke tega števila, npr. $f(3754143) = 7$. Z uporabo logaritmov izračunaj $f(f(2024^{2025}))$.

[3t] 1

$$f(x) = \lfloor \log_{10} x \rfloor + 1$$

$$\cancel{f(x)}$$

$$y = \cancel{f(2024^{2025})} = \lfloor (\log_{10} 2024) \cdot 2025 \rfloor + 1 =$$

$$f(y) = \lfloor \log_{10} (\lfloor \log_{10} 2024 \cdot 2025 \rfloor + 1) \rfloor + 1 =$$

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